

DOMAIN DRIVEN DESIGN (DDD)

IMPROVING COMMUNICATION BETWEEN
TECHNICAL AND NON-TECHNICAL MIND

PRESENTERS:
TERRI (TERASA) SIEBRANDS

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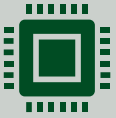
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AGENDA

- 01** Core Principles of Driven Design (DDD)
- 02** Microservices: Strategic Design Concepts
- 03** Ubiquitous Language: A shared vocabulary between technical and non-technical mind
- 04** DDD: Discover User Journeys with Storyboarding
- 05** Storyboarding: Step by Step Guide

DRIVEN DESIGN (DDD)

CORE PRINCIPLES OF DDD



Focus on the Domain

The "domain" is the sphere of knowledge and activity around which the application logic revolves. DDD emphasizes modeling this domain accurately.



Collaboration with Domain Experts

Developers work closely with business stakeholders to ensure the software reflects real-world processes and rules.



Ubiquitous Language

A shared language between developers and domain experts is used consistently in code and conversations to avoid misunderstandings.

MICROSERVICES

STRATEGIC DESIGN CONCEPTS

PROAG EXAMPLE

DOMAIN: MPCFI CLAIMS

MICROSERVICE (FEATURE): MPCFI RATING

MICROSERVICE (FEATURE): NOTICE OF LOSS (NOL)

A **Microservice** is:

- » A **semantic boundary** where a specific model applies.
- » A **contextual limit** within which terms, rules, and logic are consistent.
- » A way to **avoid ambiguity** when the same word means different things in different parts of the business.
- » We can think of a bounded context as a subdomain or modular feature of its domain

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Ubiquitous Language

A shared vocabulary between technical and non-technical mind

Ubiquitous Language is a **shared vocabulary** that is:

Developed collaboratively by domain experts and developers.

Used consistently across all forms of communication—code, documentation, conversations, diagrams, etc.

Tied directly to the domain model, meaning the terms used in the language reflect the actual business concepts and rules.

Key Characteristics

Consistency Across Boundaries

- Everyone—developers, testers, analysts, and business stakeholders—uses the same terms to describe the same concepts.

Reflected in Code

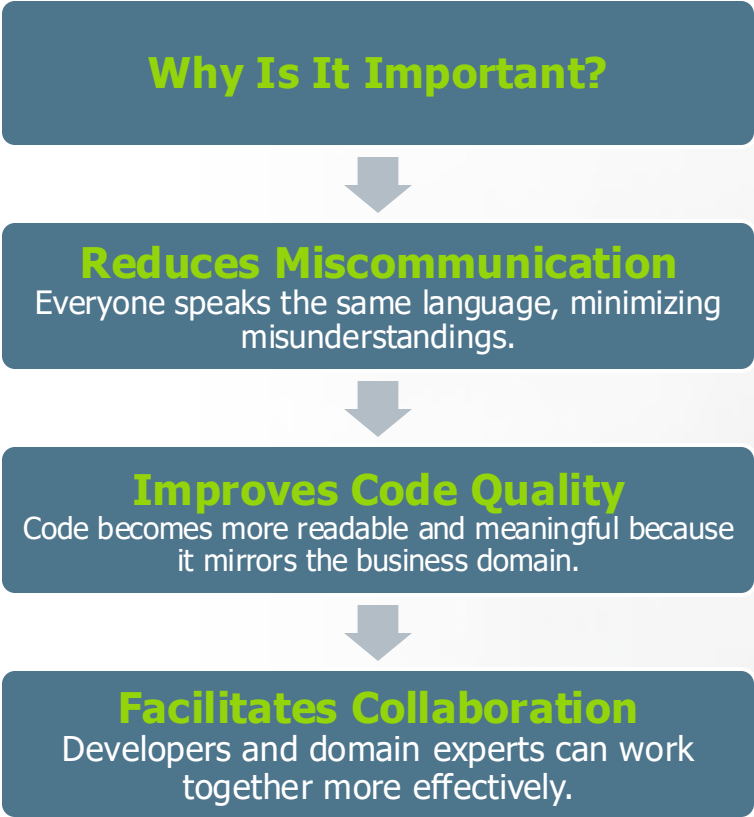
- The language isn't just for meetings or documentation. It appears in class names, method names, and variable names.

Evolves with Understanding

- As the team learns more about the domain, the language evolves to reflect deeper insights.

UBIQUITOUS LANGUAGE

DOMAIN EXAMPLES



Term	Definition	Example Usage
Policyholder	The individual or entity who owns the insurance policy.	The policyholder must submit a claim within 30 days of the incident.
Premium	The amount paid for an insurance policy, typically monthly or annually.	She pays a \$150 monthly premium for her health insurance.
Claim	A request made by the policyholder for payment based on the terms of the policy.	He filed a claim after his car was damaged in an accident.
Deductible	The amount the policyholder must pay out-of-pocket before the insurance company pays a claim.	Her policy has a \$500 deductible for medical expenses.
Coverage	The extent of protection provided by an insurance policy.	The homeowner's policy includes coverage for fire and theft.
Beneficiary	The person or entity designated to receive the insurance payout.	His wife is the beneficiary of his life insurance policy.
Adjuster	An insurance professional who investigates and evaluates claims.	The adjuster visited the site to assess the damage.

STORYBOARDING

GETTING STARTED

RECOMMENDED ROLES



- **Facilitator** - Runs the workshop or meeting, focusing on interacting with participants and keeping things flowing and on-schedule



- **Scribe** - Person who makes real-time updates to the board to reflect the current consensus to everyone.



- **Team** - Everyone else who work together to create the event storm for a specific user experience.

AGENDA

2-Hour Workshop Agenda



- Welcome, Introductions and Overview**
15 mins



- Refine the Storyboard**
40 mins

WHAT IS STORYBOARDING?



STORYBOARDING

THE LEGEND

Domain Event

Domain Event: An event that occurs in the business process written in past tense.

External System

External System

A third-party service provider such as a payment gateway or shipping company.

User/Actor

User/Actor: Someone who interacts with the business process, such as customers, partners or employees.

Command

Command

A command executed by a user through a view on an aggregate that results in the creation of a domain event.

Notes

Notes

Any additional details.

Hotspot

Hotspot: A current friction, question or clarification in the business process.

Business Process

Business Process

Processes a command according to business rules and logic. Creates one or more domain events.

STORYBOARDING

STEP 1 (INSURANCE EXAMPLE)

Collect Domain Events (Big Picture)

Each participant uses only orange Post-Its in the first round. Each orange The verb on the Post-it must therefore be in the past.

- **The first round is a pure brainstorming process about the existing domain events.**
- **Post-it:** It represents a professional event.
- **Professional event:** A technical relevant fact that happened during business.



STORYBOARDING

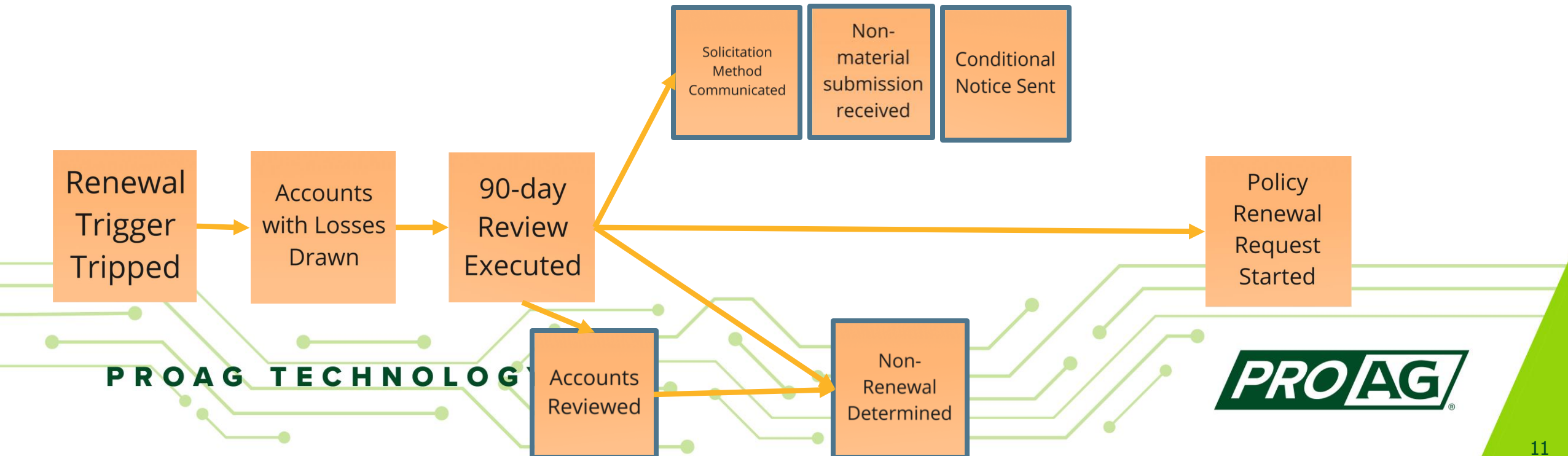
STEP 2 (INSURANCE EXAMPLE)

Refine Domain Events (Big Picture)

Ask participants to explain what each event means.

Check for syntactical correctness. Also discuss again whether the events are in the right order in terms of time. Unify occurring synonyms (different terms for the same thing) and sharpen differences if the same term was used to describe different **things**.

Go through the Domain Event Post-Its with the participants.



STORYBOARDING

STEP 3

TIMELINE CONSIDERATION

- Which users trigger which commands?
- Which commands affect which aggregates or external systems and trigger which changes?
- Which aggregates or external systems trigger which events during command processing?
- Which events trigger which policies?
- Which events create which read models for which use cases? Which policies call up which new commands?"

STORYBOARDING

STEP 4 (INSURANCE EXAMPLE)

Track causes (Process Modelling)

Where do the professional events come from? There are four main causes:

- **user actions** (Commands, User/Actors, View/Read Model)
- **external systems** (pink Post-it)
- **time** (e.g. appointment elapsed) (Business Process)
- **other domain events** (through automatic reactions).

INTERACTIVE GENERAL INSURANCE EXAMPLE

QUESTIONS?

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